

1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: _____

July 7, 2018. Saturday

Department/School: _____

SID: _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu) = \int_0^{+\infty} e^{-x} x^{\nu-1} dx, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}.$$

Select five out of following six problems.

1.(20pts.) A rod of rest length L_0 moves with a speed v along the horizontal direction, i.e., the direction of the x -axis. The rod makes an angle of θ_0 with respect to the x' -axis in the S' frame in which the rod is at rest.

(a) Show that the length L of the rod as measured by a stationary observer is $L = L_0(1 - \beta^2 \cos^2 \theta_0)^{1/2}$ where $\beta = v/c$.

(b) Show that the angle θ the rod makes with the x -axis as the stationary observer measured is determined by $\tan \theta = \gamma \tan \theta_0$ where $\gamma = (1 - \beta^2)^{-1/2}$.

2. (20pts.) A person on Earth observes two spacecrafts moving in the same direction toward the Earth. Spacecraft A has a speed of $0.50c$, and spacecraft B has a speed of $0.80c$.

(a) What is the speed v_A of spacecraft A measured by an observer in spacecraft B?

(b) What is the speed v_B of spacecraft B measured by the observer in spacecraft A?

3.(20pts.) A block of ice of mass 1 kg and at temperature of 0°C is put in thermal contact with a grand heat bath at temperature of 25°C . After some time, the ice and the heat bath are in equilibrium with each other. The latent heat of melting of ice (at 0°C) is $3.35 \times 10^5 \text{ J/kg}$ and the specific heat capacity of water at constant pressure is $4.187 \times 10^3 \text{ J/kg} \cdot \text{K}$.

- (a) Find ΔS_1 , the change in entropy of the ice (water).
- (b) Find ΔS_2 , the change in entropy of the heat bath.
- (c) Find ΔS_t , the change in entropy of the total system.

4. (20pts.) When a gas expands from volume V_0 to V at temperature T_0 , the work done by the gas is given by

$$W = RT_0 \ln \frac{V}{V_0}$$

where R is the gas constant. And the entropy of the gas at temperature T and volume V is given by

$$S = R \frac{V_0}{V} \left(\frac{T}{T_0} \right)^\alpha$$

where α is a positive constant.

(a) Write down the differential form dF , where $F = F(T, V)$ is the Helmholtz free energy of the system.

(b) Taking (T_0, V_0) as the reference point and $F(T_0, V_0) = F_0$, find the Helmholtz free energy of the system at temperature T_0 and volume V , $F(T_0, V)$.

(c) Using the condition given in (b), find the Helmholtz free energy of the system at temperature T and volume V , $F(T, V)$.

(d) Find the internal energy U , and then the heat capacity at constant volume C_V of the system.

5. (20 pts.) There is a system of gas with fixed number of molecules.

(a) Show that

$$dS = \frac{C_p}{T} dT - \left(\frac{\partial V}{\partial T} \right)_p dp$$

where S , T , p , V and C_p are the entropy, temperature, pressure, volume and heat capacity at constant pressure of the system, respectively.

(b) Show that $\left(\frac{\partial T}{\partial p} \right)_S = \frac{T}{C_p} \left(\frac{\partial V}{\partial T} \right)_p$.

(c) If the system is a monatomic ideal gas of n mole, find $(\partial T / \partial p)_S$. Will the temperature of the gas increase or decrease after a reversible adiabatic expansion? Why?

(d) For a monatomic ideal gas of n mole, show that the state of system obeys the equation

$$Tp^{\frac{2}{5}} = C$$

in the adiabatic process, where C is a constant.

6. (20pts.) In an effusive beam of an ideal gas, the number of molecules with z-component of velocity within $v_z \rightarrow v_z + dv_z$ that passes through the hole per unit area per unit time is given by

$$d\Gamma(v_z) = n_0 \left(\frac{m}{2\pi k_B T} \right)^{1/2} v_z e^{-\frac{mv_z^2}{2k_B T}} dv_z, \quad v_z > 0$$

where n_0 is the number density of molecules in the container, m the mass of the molecule, T the temperature, and k_B the Boltzmann constant. Here the normal direction of the escaping hole is assigned as the direction of the z-axis.

(a) Calculate the average of the z-component of the velocity, $\langle v_z \rangle$, of the molecules in the effusive beam.

(b) Calculate $\langle v_z^2 \rangle$ and then the uncertainty (fluctuation) $\Delta v_z = \sqrt{\langle v_z^2 \rangle - \langle v_z \rangle^2}$.

(c) Show that the molecular flux of the effusive beam, i. e., the total number of molecules that passes through the hole per unit area per unit time is given by $\Gamma = p / \sqrt{2\pi m k_B T}$ where p is the pressure of the gas in the container.